How much chloride is there in seawater?

Intended Learning Outcomes

- Know what a titration experiment is and what it is used for.
- Be able to carry out a titration experiment using silver nitrate against seawater using an indicator solution (fluorescein).
- Be able to use titration results to quantify the concentration of an analyte.

Introduction

In this experiment, you will determine the amount of chloride present in seawater using a titration. Titration is a common laboratory method for quantifying concentrations based upon finding the equivalence point of a reaction (also referred to as the end-point). The titrant (of known concentration) is added stepwise to an analyte solution until there is an indication that the reaction is complete. In practice, the indicator may be a visible change (colour, precipitate) or a change in electrical conductivity, pH or some other measurable parameter. The volume of titrant is recorded and used to calculate the analyte concentration – hence it is often referred to as a volumetric determination.

In this experiment you will be titrating silver nitrate solution (the titrant) against "seawater"[1] (the analyte) using fluorescein (an organic dye) indicator solution. The reaction is as follows:

$$AgNO_3\ (aq) + NaCl\ (aq) \leftrightarrow NaNO_3\ (aq) + AgCl\ (s)$$

Please follow the instructions carefully. In particular, note that the micro-chemical procedures that we will be using are designed to minimise the amount of chemicals used and their impact on the environment.

Apparatus and chemicals

Burette (approx. 10 ml vol) – this is your syringe used in the pH experiment.
Two 50 ml beakers
0.5M silver nitrate ($AgNO_3$) solution
Seawater solutio9n
0.2M fluorescein indicator (an organic dye)

Experimental Procedure

1. Calibrate your burette droplet size by filling with distilled water and counting drops equivalent to 0.5 ml (or use the calibration done for the pH experiment)

   There are 0.04 ml/drop based on the calibration for the pH-conductivity experiment

2. Once your burette is calibrated then empty it of any remaining distilled water.
3. Now fill the burette with **no more** than 3 ml of silver nitrate solution.
4. Using another syringe add 2.5 ml of "seawater" solution to a 10 ml beaker.
5. Add 2-3 drops of fluorescein indicator to the beaker.
6. Titrate the seawater with silver nitrate solution. You may decide to add 1 ml directly and then count drops to obtain an accurately determined volume, it is up to you to decide, however you will need to slow down the rate of titration as you approach the

equivalence point. Overall, ensure you add small amounts at a time and swirl the flask regularly.
7. With the progress of the titration, you will observe a precipitate forms and coagulates.
8. The end point is indicated by a sudden development of colour change on the precipitate.
9. Record the volume of titrant.

Results and calculations

- Look-up how much chloride there is in seawater, record this in units of g/L. You will compare this to the value you obtain from your results.

  Approximately 19.4 g/L Cl in natural seawater.

- What is the white precipitate that you see during the titration?

  Silver chloride

- Calculate the number of moles of silver nitrate used in the titration (assume silver nitrate in solution fully dissociates into its constituent ions).

  n (moles) = M (moles/L) x V (L)
  So if the volume of silver nitrate used is e.g. 2.7 ml then
  n = 0.5 x (2.7/1000) = 1.35 x 10^{-3} moles

- Find the number of moles of chloride ions in the seawater sample in 1 ml, and then in in g/L to compare with the value you looked-up earlier. Comment on how they compare. The atomic mass of Cl = 35.453.

  According to the reaction equation: moles Ag^+ = moles Cl^- = 1.35 x 10^{-3} moles
  The concentration of Cl^- in 2.5 ml of "seawater" is
  M (mol/L) = n (moles) x V(L) = 1.35 x 10^{-3}/(2.5/1000) = 0.54 mol/L
  Convert moles to grams of Cl^- by multiplying by the atomic mass of Cl = 35.453
  This gives 0.54 x 35.453 = 19.1 g/L similar to the value looked up.

  Note that in our experiment AgCl is a solid precipitate. We could have filtered, dried and then weighed the mass of this precipitate as another means to determine the mass of chloride. This kind of experiment, based on mass of precipitate, is known as a gravimetric analysis. A careful analyst would obtain the same result using either volumetric or gravimetric methods.